Datasheet

ESP32 MiniPLC



Part No.: MiniPLC-R1

Manufacturer: ISYSTEMS AUTOMATION S.R.L.

Hardware Version: V1.0

Architecture: Modular (MCU Board + RelayBoard + USB Board)

Controller: ESP32-WROOM-32U

Functional Overview

The HomeMaster MiniPLC is a professional, open-source DIN rail controller built around the ESP32 platform, designed for robust and scalable smart automation in residential, commercial, and light industrial environments. It combines extensive onboard I/O—including relays, digital and analog inputs, temperature sensors, and a user interface—with native Home Assistant integration via pre-installed ESPHome.

Engineered for reliability and flexibility, the MiniPLC supports multiple power input options (24V DC or wide-range AC/DC) and features a non-isolated RS-485 (MAX485; transceiver shares logic ground) Modbus interface for expansion with a wide range of compatible I/O modules. Its local processing capability ensures continued operation even when network or cloud connectivity is lost, making it suitable for mission-critical applications.

This controller operates as a complete standalone automation system using its comprehensive onboard I/O, while also offering seamless expansion via the RS-485 bus to connect HomeMaster smart modules for energy metering, lighting control, security, and more. It provides a versatile, ready-to-integrate platform for HVAC control, energy management, security systems, lighting automation, and custom industrial applications.

The system is implemented using three hardware boards: MCU board, USB board and Relay board.

Board	Function
MCU Board	CPU, analog acquisition, Ethernet, USB
Relay Board	Power supply, relays, digital inputs
USB Board	USB-C and SD storage

Technical Specifications

Specification	Details
Microcontroller	ESP32-WROOM-32U-N16 (dual-core)
Storage	MicroSD card (SPI interface, power-switched 3.3 V rail)
Power Input	24 V DC nominal or 85–265 V AC, 47–63 Hz (L / N)
RTC	RTC (PCF8563); backup battery holder, battery not fitted
Document revision	DS-MiniPLC Rev. B · 2026-09 · Hardware V1.0
Digital Inputs	4 × IEC 61131-2 compliant 24 V digital inputs (ISO1212 front-end) (DIx + GNDx), dry-contact (module-wetted), not galvanically isolated; per-channel PTC fuse, TVS surge suppression, reverse-polarity protection, EMI filtering
Relay outputs	6 × SPDT (NO / C / NC); 3 A @ 250 VAC per contact (module limit)
Analog Inputs	4 × 0–10 V, 16-bit (ADS1115)
Analog Outputs	1 × 0–10 V, 12-bit (MCP4725)
Temperature Inputs	2 × RTD (PT100/PT1000 via MAX31865), 2 × 1-Wire (DS18B20 compatible)
Display	128 × 64 OLED (SH1106)
User Interface	4 buttons, 16 LEDs: PWR ×1, user ×3 (U.1–U.3; U.3 is the status LED by default, reassignable), RX/TX ×2, DI ×4, relay ×6
RS-485	non-isolated RS-485 (MAX485; transceiver shares logic ground); half-duplex Modbus RTU, not galvanically isolated
Wi-Fi	Wi-Fi (ESP32)
Ethernet	Optional Ethernet (LAN8720 PHY)
USB	USB-C (ESD protected, CC detection, data to ESP32)

Top Terminal Block (1–28)

Pos	Marking	Signal	Description
1	L	POWER_AC	AC line (MAINS)
2	N	POWER_AC	AC neutral (MAINS)
3	V+	POWER_DC	24 V DC input
4	0V	POWER_DC	24 V DC return
5	+5V	BUS1	1-Wire bus 1 supply
6	D	BUS1	1-Wire bus 1 data
7	Gnd	BUS1	1-Wire bus 1 ground
8	+5V	BUS2	1-Wire bus 2 supply
9	D	BUS2	1-Wire bus 2 data
10	Gnd	BUS2	1-Wire bus 2 ground
11	NO	RELAY1	Relay 1 normally

			open
12	C	RELAY1	Relay 1 common
13	NC	RELAY1	Relay 1 normally closed
14	NO	RELAY2	Relay 2 normally open
15	C	RELAY2	Relay 2 common
16	NC	RELAY2	Relay 2 normally closed
17	NO	RELAY3	Relay 3 normally open
18	C	RELAY3	Relay 3 common
19	NC	RELAY3	Relay 3 normally closed
20	NO	RELAY4	Relay 4 normally open
21	C	RELAY4	Relay 4 common
22	NC	RELAY4	Relay 4 normally closed
23	NO	RELAY5	Relay 5 normally open
24	C	RELAY5	Relay 5 common
25	NC	RELAY5	Relay 5 normally closed
26	NO	RELAY6	Relay 6 normally open
27	C	RELAY6	Relay 6 common
28	NC	RELAY6	Relay 6 normally closed

Bottom Terminal Block (1–26)

Pos	Marking	Signal	Description
1	GND	ANALOG_INPUT	Common return for AI1-AI4
2	AI1	ANALOG_INPUT	Analog input 1, 0-10 V
3	AI2	ANALOG_INPUT	Analog input 2, 0-10 V
4	AI3	ANALOG_INPUT	Analog input 3, 0-10 V
5	AI4	ANALOG_INPUT	Analog input 4, 0-10 V
6	AO	ANALOG_OUTPUT	Analog output, 0-10

			V
7	GND	ANALOG_OUTPUT	Analog output return
8	1	RTD1	RTD 1 terminal 1
9	2	RTD1	RTD 1 terminal 2
10	3	RTD1	RTD 1 terminal 3
11	4	RTD1	RTD 1 terminal 4
12	1	RTD2	RTD 2 terminal 1
13	2	RTD2	RTD 2 terminal 2
14	3	RTD2	RTD 2 terminal 3
15	4	RTD2	RTD 2 terminal 4
16	GND	DI1	DI1 return
17	I.1	DI1	DI1 input
18	GND	DI2	DI2 return
19	I.2	DI2	DI2 input
20	GND	DI3	DI3 return
21	I.3	DI3	DI3 input
22	GND	DI4	DI4 return
23	I.4	DI4	DI4 input
24	COM	RS485	RS-485 signal reference
25	B	RS485	RS-485 data -
26	A	RS485	RS-485 data +

*Relay outputs are **not internally fused**. External overcurrent protection must be provided in accordance with applicable standards and installation requirements. Use appropriately rated fuses or circuit breakers and consider external contactors for loads exceeding 3 A or for inductive and high inrush loads.

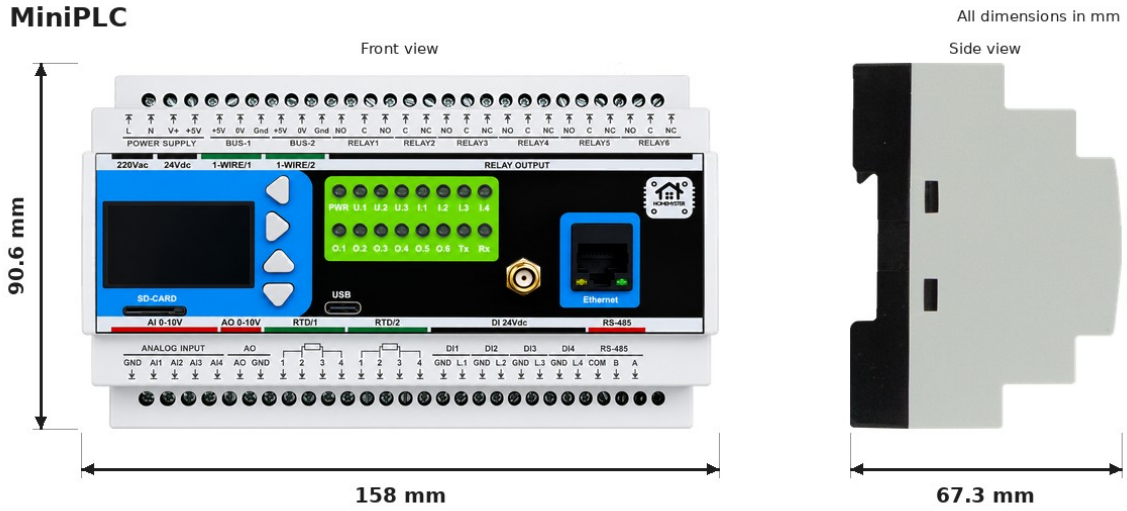
Installation, Environmental & Mechanical

Category	Specification	Details
Environmental Ratings	Operating temperature	0 °C to +40 °C
Environmental Ratings	Storage temperature	−10 °C to +55 °C
Environmental Ratings	Relative humidity	0–90 % RH, non-condensing
Environmental Ratings	Ingress protection	IP20 (inside cabinet only)
Environmental Ratings	Installation	Indoor control cabinet only; not for outdoor or exposed installation
Environmental Ratings	Maximum altitude	2000 m
Environmental Ratings	Pollution degree	2
Mechanical & Packaging	Dimensions	158 × 90.6 × 67.3 mm (L × W × H)
Mechanical & Packaging	DIN width	9 modules (≈ 158 mm)
Mechanical & Packaging	Mounting	35 mm DIN rail
Mechanical & Packaging	Enclosure	PC/ABS industrial enclosure

Mechanical & Packaging	Terminal type	Pluggable screw terminal blocks, 5.08 mm pitch
Mechanical & Packaging	Wire cross-section	0.2–2.5 mm² (AWG 24–12)
Mechanical & Packaging	Tightening torque	0.4–0.6 Nm
Mechanical & Packaging	Net weight	300 g
Mechanical & Packaging	Gross weight	450 g
Mechanical & Packaging	Pack size	230 × 140 × 87 mm (L × W × H)

Install only inside a control cabinet with ventilation; the cabinet must include a protective front plate covering all module connection terminals and a closing protective door; not for outdoor or exposed installation.

All wiring terminals must be protected against accidental contact by an insulating front plate, wiring duct, or terminal cover. Exposed live terminals are not permitted.



Cable Recommendations & Shield Grounding

This section applies to **Analog (0–10V)**, **Temperature (RTD / 1-Wire)**, and **RS-485**. Use shielded, twisted constructions and bond shields correctly to reduce EMI and ground-loop issues.

General Routing Rules

- Route low-level signal cables (Analog/RTD/1-Wire/RS-485) separately from mains, relay outputs, contactors, VFD motor cables, and power wiring.
- If crossing power cables is unavoidable, cross at **90°**.
- Keep cable runs as short as practical and avoid parallel runs with high-current conductors.

Analog (0–10V) Cable

- **Construction:** twisted pair (Signal + GND) per channel
- **Shielding:** overall shield (standard) or individually shielded pairs (high-EMI)
- **Examples: J-Y(ST)Y** (overall shield) or **LI2YCY PIMF** (shielded twisted pairs; one pair per channel)

Temperature Cable

RTD (PT100/PT1000)

- **Recommended:** shielded multi-core for 2/3-wire; shielded pairs for best accuracy (4-wire)
- **Examples: J-Y(ST)Y** (overall shield) or **LI2YCY PIMF** (pairs; e.g. 2×2×0.50 for 4-wire)

1-Wire (DS18B20)

- **Recommended:** shielded 3-core (+5V / DATA / GND) for typical installations
- **High-EMI / long runs:** shielded pairs + overall shield (e.g. **LI2YCY PIMF 2×2×0.50**)
- **Topology:** daisy-chain (bus). Avoid star wiring.
- **Stubs:** keep sensor stubs ≤ **0.5 m**.
- **Pull-up (DATA): 4.7 kΩ** typical; **2.2–3.3 kΩ** for long/heavy loads.

RS-485 Cable

- **Construction:** twisted pair for **A/B**
- **Characteristic impedance: 120 Ω** (recommended)
- **Shielding:** overall shield is recommended in cabinets; use individually shielded pairs + overall shield in high-EMI
- **Examples: J-Y(ST)Y 2×2×0.5 mm²** or **LI2YCY PiMF 2×2×0.50**
- Use one twisted pair for A/B. Run COM to every node. Required, not optional: the RS-485 port is not galvanically isolated.

Shield Grounding

- **Default recommendation:** bond cable shield(s) to cabinet **PE/EMC ground at the PLC end only**.
- **Do not** connect shields to signal terminals (AI/AO/RTD/1-Wire/RS-485 A/B/COM).
- If both ends are in equipotential bonded cabinets, shields may be bonded at both ends using proper 360° clamps.

Compliance & Certifications

The HomeMaster MiniPLC is CE marked and designed to comply with the applicable European Union directives. The manufacturer, ISYSTEMS AUTOMATION (HomeMaster brand), maintains the technical documentation and a signed EU Declaration of Conformity (DoC).

EU Directives

EMC 2014/30/EU · LVD 2014/35/EU · RED 2014/53/EU · RoHS 2011/65/EU

Harmonised Standards

Area	Standards
EMC	EN 61000-6-1 (Immunity) · EN 61000-6-3 (Emissions)
Electrical Safety	EN 62368-1
Radio	EN 300 328 · EN 301 489-1 · EN 301 489-17
RoHS	EN IEC 63000

Radio

The product integrates a pre-certified ESP32 Wi-Fi radio module (2.4 GHz). Final product conformity with the Radio Equipment Directive is demonstrated by the maintained technical documentation and conformity assessment of the complete device.

Safety notice

L / N: hazardous voltage · 24 V DC: SELV · Qualified personnel only

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